



Divisibility tests for 2 and 3

Task A: Any even number is a multiple of 2

- 1) Using a test of divisibility by 2, identify all the multiples of 2 from the list below:

3 284 738

2 478 349

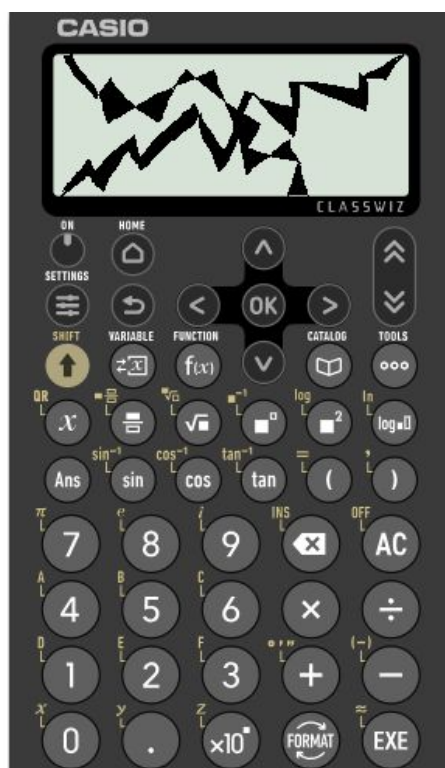
999 111 333

445 534

390

1115

222 222 222



- 2) Izzy's calculator has broken.

She knows the number on the screen :

is a multiple of 2

8650

has exactly 5 tens

8652

is a 4 digit number

8654

is greater than 8600

8656

Less than 8660

What could her number be? 8658

Task B: Divisibility tests for multiples of 3

1a) Identify all the multiples of 3 from the list.

234,

1398,

20 495,

593,

56 576

$$2+3+4=9 \quad 1+3+9+8=21 \quad 2+4+9+5=20 \quad 2+9+3=17 \quad 5+6+5+7+6=29$$

1b) Identify all the multiples of 2 and 3 from the list.

325,

139,

120 490,

594,

656 578 452

$$1+2+4+9=16 \quad 5+9+4=18 \quad 6+5+6+5+7+8+4+5+2=48$$

2) Here are five cards, each with a digit.



Only using 4 cards,

a) make a 4 digit number which is a multiple of 3?.

*Any four digit number using the digits 1, 1, 4 and 6.
Example 1146, 6114 or 4161*

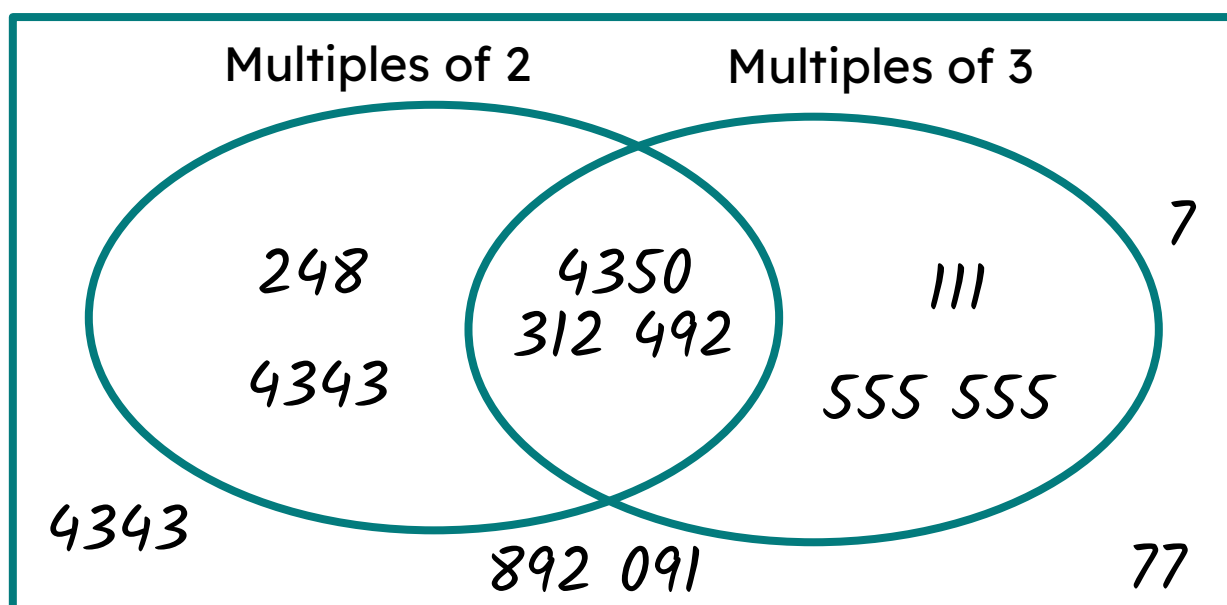
b) make a 4 digit number which is a multiple of 3 and 2.

*Any four digit number using the digits 1, 1, 4 and 6 with 4 or 6 in the ones column.
Example 1146, 6114 or 1614*

Task B: Divisibility tests for multiples of 3

3) Using the numbers below, fill in the Venn diagram.

248 4343 4350 312 492 892 091
111 7 77 555 555



4) Here is an integer with a lot of digits.

999 898 969 698 751 986

Using the divisibility test, how do you show this is a multiple of 3?

$$9+9+9+8+9+8+9+6+9+6+9+8+7+5+1+9+8+6 = 135$$

$$1+3+5 = 9$$

999 898 969 698 751 986 is divisible by 3

